Life Care Pharmacy
Online Medical Store Website

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BACHELOR OF SCIENCE IN COMPUTER SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

Government College University

Faisalabad

Life Care Pharmacy Online Medical Store Website

BY

Zara Asmat Ullah 2021-GCUF-085702

Project submitted by the fulfilment of the requirement for the degree of

BACHELOR DEGREE IN
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DEPARTMENT OF COMPUTER SCIENCE

GOVERNMENT COLLEGE UNIVERSITY FAISALABAD

DECLARATION

I so certify that this project report, "Life Care Pharmacy Website," is entirely original and was finished for educational purposes. I am the author and organizer of all the information contained in this documentation.

All materials and references utilized in this project have been duly credited.

I further declare that no other institution or group has received this project for any other reason.

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PROJECT COMPLETION REPORT

Title of the project: Life Care Pharmacy
Online Medical Store Website

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It is to certify that the title & proposal submitted by the students are clear and according to project scope.

It is further certifying that the project of the students may kindly be process for external examiner/viva voice.

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CERTIFICATE BY THE PROJECT SUPERVISOR

This is to certify that the project titled **“Development of an Online Medical Website”** has been completed by **Zara Asmat Ullah** under my supervision **Prof. Muhammad Kamran** as a partial fulfillment of the requirements for the degree/program of **BS (Computer Science)**

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I also want to express my gratitude to my instructors, friends, and everyone else who helped me finish my project, whether directly or indirectly.

DEDICATION

First and foremost, I want to sincerely thank Almighty Allah for his blessings, mercy, and guidance, which enabled me to successfully finish this assignment.

I dedicate this effort to my dear mother and sisters in appreciation of their unending prayers, care, support, and encouragement during my journey. Their drive and selflessness have always motivated me to put in a lot of effort and accomplish my objectives.

I also dedicate this endeavor to my esteemed instructors for their invaluable advice and assistance throughout the project's completion.

Lastly, this project is devoted to my educational journey, which has given me information, real-world experience, and self-assurance in my abilities.

May Allah bless you.

Ameen

Table of Contents

Chapter
Introduction

1. Introduction

This project is a web-based application created to enhance the management and accessibility of pharmacy and medical services. It offers a digital platform that eliminates the need for customers to physically visit a store in order to examine available medications, check details, and place purchases. For both users and administrators, the system seeks to streamline, expedite, and organize the process.

People frequently encounter issues with traditional approaches, including lengthy wait periods, lack of access to medications, and time consumption. By offering an online solution that streamlines the entire procedure and increases accessibility to healthcare-related products, this project tackles these issues.

1.1. PROBLEM STATEMENT

Customers have many challenges while trying to obtain medications and medical supplies via the conventional drugstore system. It takes time and effort for them to physically visit stores. Patients frequently experience treatment delays and annoyance, when **necessary,** medications are not readily available. In pharmacies, manual order processing and record keeping can also result in mistakes, poor supply management, and trouble tracking consumer orders. Additionally, there isn't a centralized system that makes it simple for customers to compare products, verify availability, and place orders effectively.

A digital solution that offers a simple, quick, and dependable platform for managing pharmacy-related services online is required to address these problems.

1.2. Objectives of the Project

This project's primary goal is to create a website for Lifecare Pharmacy in order to enhance and update the pharmacy's offerings. The goal of this website is to give users a simple and effective way to access pharmacy-related services and information online.

The project's particular goals are:

- To create a responsive and easy-to-use website.
- To make **medications** and pharmacy services accessible online.
- To enhance communication between the pharmacy and customers.
- To digitize pharmaceutical procedures in order to minimize manual labor.
- To provide quick and simple access to product availability and information.
- To make it more convenient for customers to browse products online at any time.
- To develop a dependable and safe method for handling pharmacy data.

1.3. Scope of the Project

The project's goal is to create a website for lifecare Pharmacy that offers an online platform for pharmacy service management and access. By facilitating information access and enhancing service effectiveness, the system is intended to assist both customers and pharmacy employees.

This website has features like showcasing medications and health supplies, offering comprehensive product details, and making it simple for consumers to peruse the items that are available. Additionally, it lessens the need for manual record keeping and aids in the orderly organization of pharmacy data.

The project's scope includes essential e-commerce features tailored for an online pharmacy system. It allows customers to browse available medicines, manage products in a shopping cart, upload prescriptions, and securely place orders using the Cash on Delivery (COD) payment method.

Chapter 2
SYSTEM ANALYSIS

2.0 System Analysis:

The LifeCare Pharmacy website project was created to examine and address issues with conventional pharmacy administration systems. Customers' access to medication information is restricted under the manual approach, and pharmacy employees spend more time maintaining records and responding to client inquiries. This initiative offers a digital solution to increase accessibility and efficiency.

An online pharmacy website can make it simple for consumers to search for and examine medications, health goods, and associated information from any location, according to the project's analysis. Additionally, it lessens manual labor and enhances customer-pharmacy communication.

User requirements, usability, functionality, and performance were taken into consideration when analyzing the system. Users may effortlessly navigate through various sections of the website because to its straightforward and userfriendly interface. The project also prioritizes systematic data management and precise product information.

The following crucial needs were discovered during the analysis stage:

- A user-friendly and responsive website design.
- Information on medications and products is easily accessible.
- Pharmacy data should be managed and organized properly.
- Quick and effective system operation.
- Safe management of pharmacy and user data.

2.1. Existing System

The majority of the current pharmaceutical administration system is manual and paper-based. Customer records, medication information, and stock management are all done by hand in traditional pharmacies, which requires more time and effort. To learn more about medications and services, customers must physically visit the pharmacy.

Issues with the Current System

- Customer information and medication records are manually managed.
- Greater paperwork and data loss risks.
- Searching for medical information is a time-consuming task.
- Restricted access to product information and pharmacy services.
- Absence of an internet platform for clients.
- Managing and updating stock information can be challenging.
- Customers and pharmacy employees communicate slowly.
- Increased likelihood of human error in record keeping.
- Inadequate computerized system for pharmacy data organization.
- Users have limited access to pharmacy information.

The current method is time-consuming, inefficient, and challenging to administer. To increase productivity, accessibility, and service quality, a computerized and online pharmacy management system is required, such as the LifeCare Pharmacy website.

2.2. Proposed System

The suggested system is a web-based pharmacy management system designed to enhance and update pharmacy services for LifeCare Pharmacy. Customers don't need to physically visit the pharmacy to examine medications, medical supplies, and pharmacy-related information thanks to the system's online platform. The suggested approach is intended to replace the conventional manual system with an effective and digital one. It facilitates better customer-pharmacy communication while maintaining pharmacy data in an orderly and safe manner.

Characteristics of the Suggested System

- Responsive and easy-to-use website design.
- Prescription drugs and medical supplies are displayed online.
- Users can easily navigate and search.
- Quick access to product availability and details.
- Systematic administration of pharmacy data and records.
- Decrease in human mistake and manual paperwork.
- Enhanced interaction between drugstore employees and consumers.
- A dependable and safe approach for managing data.
- Improved accessibility with a mobile-friendly UI.
- Online pharmacy services and information are available around-the-clock.

2.3 Feasibility Study

The LifeCare Pharmacy website project's feasibility study was carried out to ascertain whether the suggested method is workable, effective, and advantageous for pharmacy administration. In terms of technology, cost, operation, and usability, the study demonstrates that the system is viable.

Feasibility Study Types:

1. Technical Viability

Because the necessary tools, software, and technologies are readily available, the project is technically feasible. Modern web technologies like HTML, CSS, JavaScript, PHP, and MySQL can be used to create the website. Standard computers with internet access can operate the system effectively.

2. Financial Viability

Because the website's creation and maintenance costs are reasonable, the project is financially feasible. Long-term operating cost savings can result from the online system's reduction of paperwork, physical labor, and time consumption.

3. The viability of operations

Customers and pharmacy employees can easily operate and maintain the suggested system. Online pharmacy services and medication information are readily available to users. The solution boosts productivity and enhances day-to-day pharmacy operations.

4. Schedule Viability

Because the necessary resources, technologies, and development process are controllable and well-planned, the project can be finished in the allotted time.

Feasibility Study Conclusion

The LifeCare Pharmacy website idea is feasible, economical, easy to use, and technically feasible, according to the feasibility study. As a result, the suggested system can enhance pharmacy management and services.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.0 Requirement Specification

A crucial stage of software development is requirement specification, which outlines the system's requirements and expectations. It outlines the actions and performance that the system should do to satisfy user needs. Before beginning the actual design and execution, this stage aids developers in comprehending the primary characteristics, capabilities, and constraints of the system.

3.1 Hardware Requirements

The physical components required for the development, operation, and testing of the LifeCare Pharmacy website project are included in the hardware requirements. These specifications guarantee the website operates correctly and effectively.

- A laptop or PC for system implementation and development.
- 4GB of RAM is the minimum; 8GB is advised for optimal performance.
- For seamless processing, use an Intel Core i3 or higher processor.
- A minimum of 500GB of hard drive space for databases and project files.
- A reliable internet connection for testing and online access.
- Common input devices, like a mouse and keyboard.
- Ahigh-resolution monitor for development work and clear display.

3.2 Software Requirements

The programs and tools needed to create, operate, and maintain the system are included in the software requirements for the LifeCare Pharmacy online project. These software solutions aid in creating a seamless, effective, and userfriendly website.

- Operating System: Linux or Windows 10 or later.
- Frontend technologies include JavaScript, HTML, and CSS.
- PHP (or any server-side language utilized in the project) is the backend language.
- MySQL is the database management system used to store and handle pharmacy data.
- Local Server: To execute the project locally, use XAMPP or Apache Server.
- Web browser: Mozilla Firefox or Google Chrome for use and testing.
- Code Editor: To write and edit code, use Visual Studio Code or Sublime Text.

3.3 Functional Requirements

The primary functions and services that the system must provide are specified in the functional requirements of the LifeCare Pharmacy website project. These specifications outline how the system will satisfy user and pharmacy management demands.

- Users should be able to view medications and medical supplies online using the system.
- The product's name, description, price, and availability should all be shown on the website.
- It should be simple for users to look for medications.
- Products should be categorized by the system to improve navigation and browsing.
- The website should have an easy-to-use interface.
- It should be possible for the administrator to add, edit, and remove product details.
- Users should be able to access pharmacy information at any time through the system.
- All services should be quickly and easily accessible via the website.

3.4 Non-Functional Requirements

The LifeCare Pharmacy website project's non-functional requirements specify how the system should operate rather than what it should accomplish. These specifications guarantee the system's dependability, performance, and quality.

- For every user, the system should be quick and responsive.
- Even with a lot of material, the page should load swiftly.
- To safeguard user and pharmacy data, the system needs to be secure.
- The website must be simple to use and navigate.
- The system must be dependable and accessible around-the-clock.
- The design ought to be responsive and compatible with desktop, tablet, and mobile devices.
- In the future, the system should be simple to upgrade and maintain.
- The website should be able to support many users without experiencing any performance problems.
- For a better user experience, the interface should be straightforward and consistent.
- Data integrity and correctness should always be guaranteed by the system.

CHAPTER 4

System Design

4.0 SYSTEM DESIGN

The Life Care Pharmacy website's system design was created to give customers an easy, safe, and effective online medical service experience. This system's primary goals are to guarantee
The front-end and back-end of the website operate independently but are completely integrated thanks to its clientserver architecture. The user interface, which enables users to browse medications, upload prescriptions, and place orders, is handled by the front-end. All system functions, data processing, and storage are handled by the back-end.

The system is made up of various crucial modules, such as:

- System for User Registration and Login.
- Medication Search and Category Management.
- Prescription Upload Function.
- Online Ordering Platform and Shopping Cart.
- Admin Panel for order, user, and medication management.
- Every piece of information, including order records, medication specifics, and user information, is safely kept in the database.
- To safeguard user data, security mechanisms such as secure transaction management, encrypted passwords, and user authentication are put in place.

4.1. SYSTEM ARCHITECTURE

The Life Care Pharmacy website's system architecture outlines how several system components work together to enable seamless operation. Presentation Layer, Application Layer, and Database Layer are the three primary layers of the client-server architecture.

1. Presentation Layer (Front-End)

This is the website's user interface. Through a web browser, users engage with this layer. It consists of:

- Homepage: Medical search page
- Pages for registration and login
- Pages for the cart and checkout
- Interface for uploading prescriptions

HTML, CSS, and JavaScript are used in the construction of this layer to guarantee a responsive and userfriendly design.

2. Application Layer (Back-End)

The system's primary logic is handled by this layer. It interacts with the database and handles user requests. It consists of:

- User verification (signup/login)
- System for managing medications
- System for processing orders
- Verification of prescriptions
- Operations of the admin panel

This layer makes sure that every feature on the website operates safely and correctly.

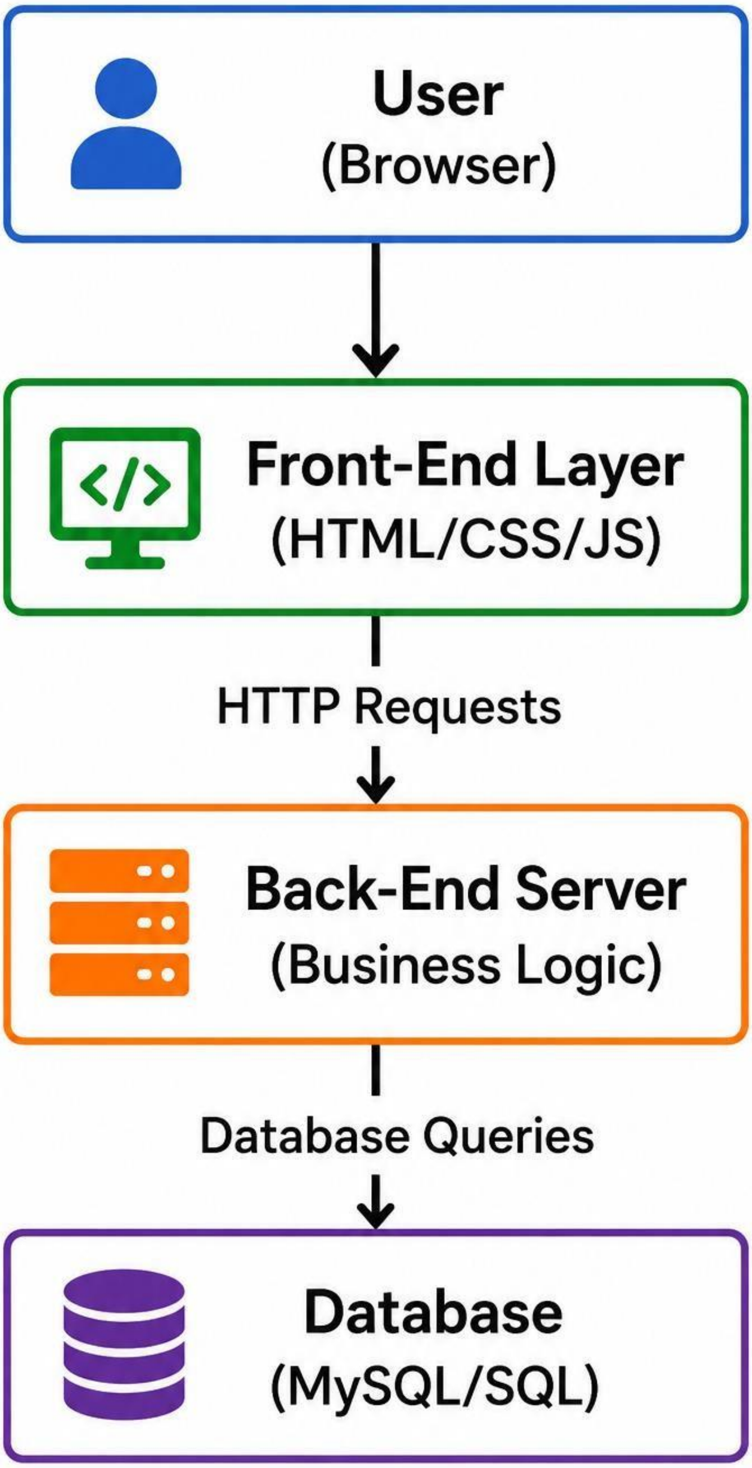
3. Database Layer

This layer contains all of the system's critical data, including:

- Details about the user
- Details about the medication
- Records of orders and transactions
- Prescription information

For safe and organized data storage, relational databases like MySQL are commonly utilized.

4.2. SYSTEM ARCHITECTURE DIAGRAM



CHAPTER 5

IMPLEMENTATION

5.0. IMPLEMENTATION

To guarantee that all system requirements are appropriately met, the online medical pharmacy website is implemented utilizing a structured development method. The system is built with a MySQL database for data storage, HTML, CSS, and JavaScript for the front end, and an appropriate back-end technology like PHP or Node.js.

5.1. DEVELOPMENT TOOLS

Modern web technologies and tools are used in the building of this online medical pharmacy website to guarantee the system's effectiveness, dependability, and seamless operation. The development process makes use of the following technologies and tools:

1.Front-End Development Tools

- The fundamental structure of web pages is created using HTML (HyperText Markup Language).
- The user interface is designed and styled using CSS (Cascading Style Sheets).
- JavaScript: Used to add dynamic features and interactivity, like cart operations, search capabilities, and form validation.
- For responsive and mobile-friendly design, use Bootstrap (optional).

2. Back-End Development Tools

Data processing and logic are handled by the back-end.

The following resources could be utilized:

- Server-side scripting and application logic are handled by PHP and Node.js.
- Express.js: Facilitates the efficient development of server-side APIs when Node.js is utilized.

3. Database Tools

For safely managing and storing data:

- MySQL: Used as a relational database to hold orders, prescriptions, medication information, and user data.
- phpMyAdmin: A graphical user interface for simple database administration.

4. Development Environment

The primary code editor for creating and organizing source code is Visual Studio Code (VS Code).

The local server environment used to test PHP and MySQL-based applications is called XAMPP or WAMP.

6. Browser Tools

- Google Chrome Developer Tools and Firefox Developer Tools: Used for website testing, debugging, and responsive design verification.

5.2. MODULES IMPLEMENTATION

To effectively manage the system, the online medical pharmacy website is separated into various parts. Each module carries out a distinct task and collaborates to offer a seamless user experience.

1. User Registration and Login Module

Users can create accounts and safely log in to the website with this module. The database contains user data, including name, email address, password, and contact information. To safeguard user accounts, authentication methods are used.

Functions:

- User signup
- User login/logout
- Password security and validation

2. Medicine Management Module

This module manages all medicine-related information. Medicines are organized into categories to make searching easier for users.

Functions:

- Add, update, and delete medicines
- Display medicine details and prices
- Search and filter medicines

3. Prescription Upload Module

This module allows users to upload medical prescriptions while ordering medicines. Uploaded files are stored securely in the database or server storage.

Functions:

- Upload prescription images/files
- Store prescription records
- Verify uploaded prescriptions

4. Shopping Cart and Order Module

This module handles the online purchasing process. Users can add medicines to the cart, update quantities, and place orders.

Functions:

- Add/remove products from cart
- Manage order details
- Confirm and track orders

5. Admin Management Module

The admin panel is implemented to control and manage the entire system. Administrators can monitor users, medicines, and orders through this module.

Functions:

- Manage medicine records
- View customer orders
- Manage users and prescriptions
- Update website content

6. Database Management Module

This module is responsible for storing and managing all website data efficiently using a relational database system.

Functions:

- Store user information
- Manage medicine and order data
- Retrieve records quickly and securely

7. Security Module

This module ensures the protection of user data and secure system operations.

Functions:

- Password encryption
- Input validation
- Secure login authentication
- Protection against unauthorized access

5.3. USER INTERFACE IMPLEMENTATION

The online medical pharmacy website's user interface is designed to give users a straightforward, appealing, and easy-to-use experience. Users may effortlessly navigate through the website's many sections thanks to the interface's design.

Main User Interface Components

1. Home Page

The home page provides an overview of the website and displays:

- Featured medicines
- Search bar
- Categories of medicines
- Navigation menu
- Promotional banners and information

The layout is designed to make important features easily accessible to users.

2. Registration and Login Pages

These pages allow users to create accounts and securely access the system. Simple forms with input validation are implemented to improve usability and security.

Features:

- User-friendly forms
- Password protection
- Error and success messages

3. Medicine Search and Product Display

The medicine section is designed to help users quickly find products.

Features:

- Search functionality
- Medicine categories
- Product images and details
- Price and availability display

4. Shopping Cart Interface

The shopping cart interface allows users to manage selected products before placing an order.

Features:

- Add or remove medicines
- Update product quantity
- Display total price
- Proceed to checkout option

5. Prescription Upload Interface

A simple upload system is implemented for users to upload prescriptions securely.

Features:

- File upload option

- Image/document support
- Upload confirmation messages

6. Admin Dashboard Interface

The admin interface is developed to manage the website efficiently.

Features:

- Manage medicines
- View orders
- Monitor users
- Update website content

Responsive Design Implementation

The website automatically adapts to various screen sizes thanks to responsive web design principles. Flexible layouts and Bootstrap grid systems enhance user experience and accessibility.

User Experience Enhancements

To improve the overall user experience, the following features are implemented:

- Easy navigation menus
- Fast-loading pages
- Attractive color scheme and layout
- Interactive buttons and animations
- Clear error and success notifications

5.4. ADMIN PANEL IMPLEMENTATION

The admin panel is used to effectively administer and control every aspect of the online pharmacy website. It gives administrators a safe way to keep an eye on users, control medications, process orders, and maintain the system as a whole.

Back-end technologies like PHP and Node.js are used in the development of the admin panel, while MySQL is used for database connectivity. To guarantee that only authorized administrators can access the panel, authentication procedures are put in place.

1. Admin Authentication System

A secure login system is implemented for administrators to protect sensitive system data and management features.

Functions:

- Admin login and logout
- Username and password verification
- Secure session management

2. Medicine Management

This module allows administrators to manage medicine records available on the website.

Functions:

- Add new medicines
- Update medicine details and prices
- Delete medicine records
- Manage medicine categories and stock availability

3. Order Management

The order management section helps administrators monitor and process customer orders efficiently.

Functions:

- View customer orders
- Update order status
- Confirm and manage deliveries
- Track order history

4. User Management

This module enables administrators to manage registered users of the website.

Functions:

- View user information
- Manage customer accounts
- Remove unauthorized or inactive users

5. Prescription Management

Administrators can review uploaded prescriptions before approving medicine orders.

Functions:

- View uploaded prescription files
- Verify prescriptions
- Approve or reject prescription-based orders

6. Dashboard and Reports

The admin dashboard provides a summary of website activities and system performance.

Features:

- Total users and orders overview
- Medicine inventory status

- Sales and order statistics
- Recent activity monitoring

7. Security and Access Control

Security measures are implemented to protect administrative data and system operations.

Features:

- Password encryption
- Role-based access control
- Secure database communication
- Protection against unauthorized access

ER Diagram – Online Medical Website

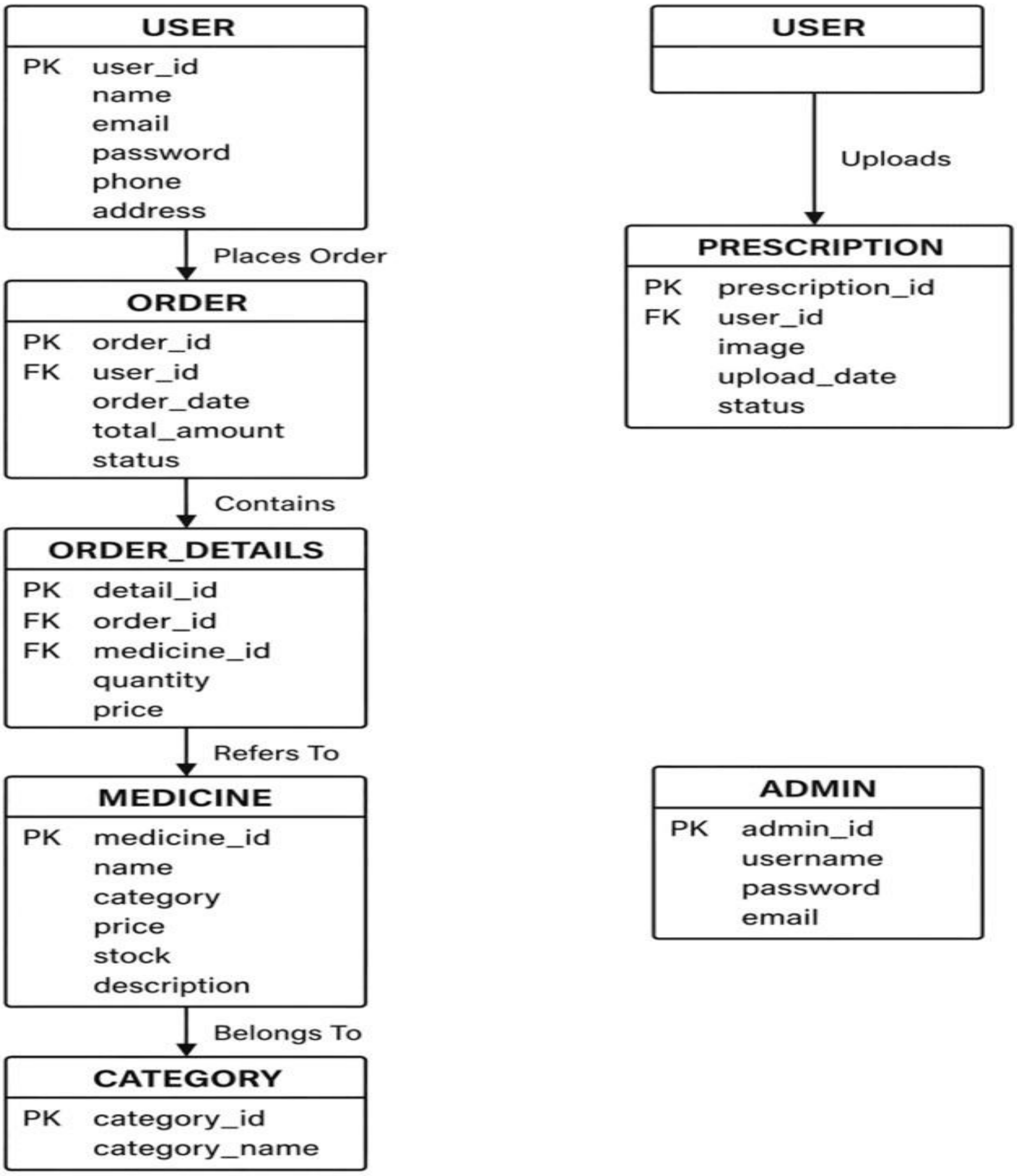


Figure 5.1: Entity Relationship Diagram (ERD)

Use Case Diagram – Online Medical Website

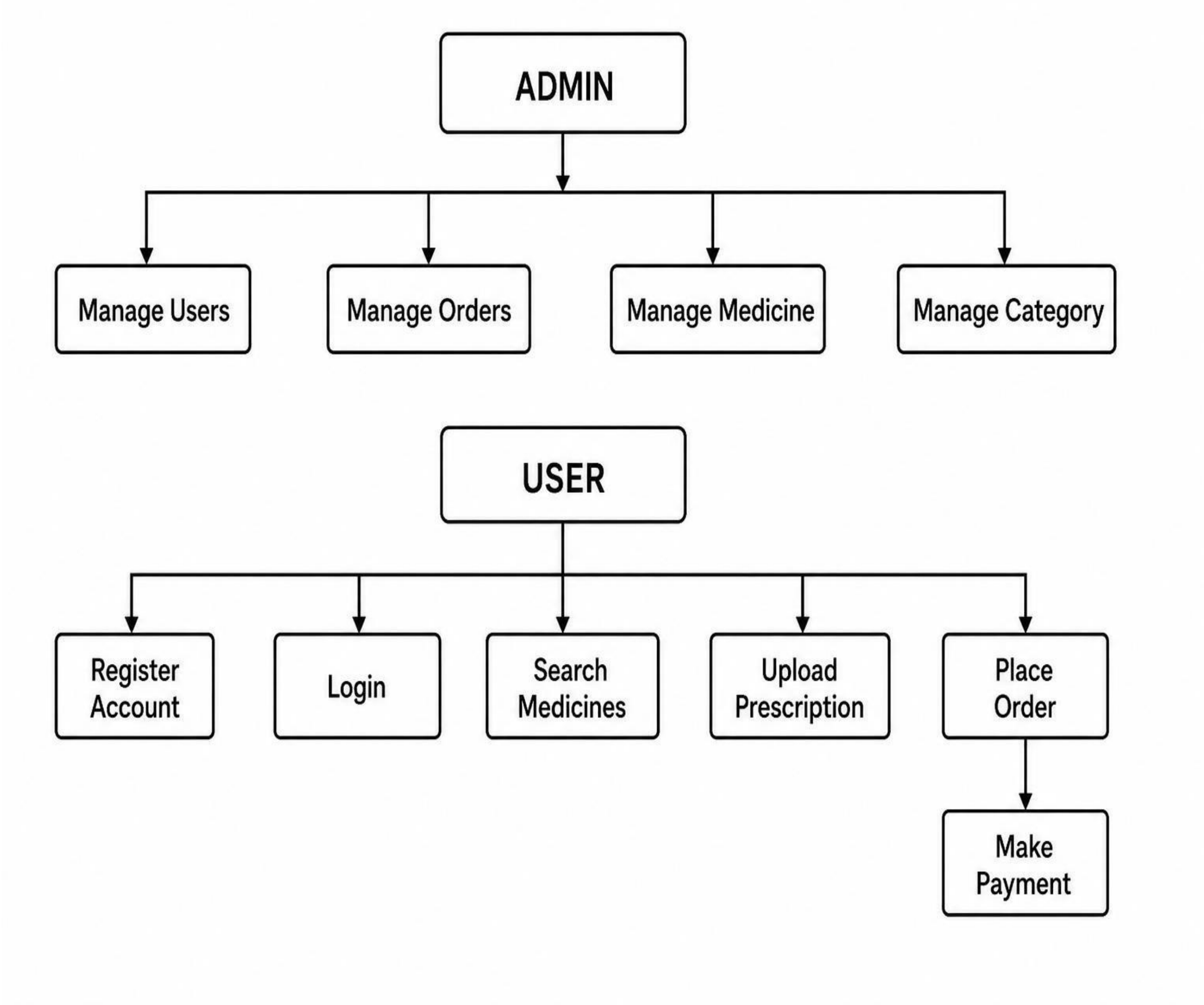


Figure 5.2: Use Case Diagram

Data Flow Diagram (DFD) – Online Medial Store

Context Level Diagram (Level 0 DFD)

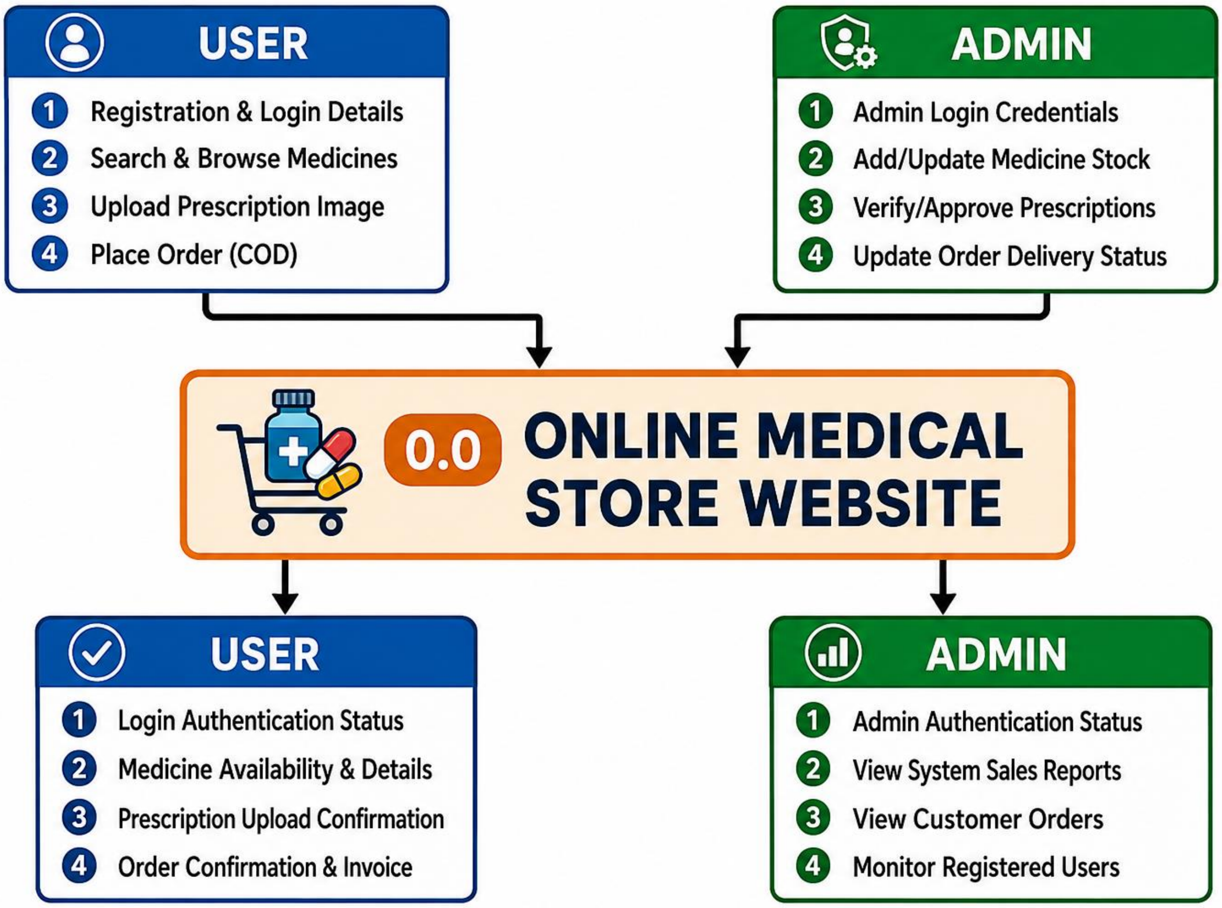


Figure 5.3: Context Level Data Flow Diagram

Description of Level 0 DFD:

The above Context Level Diagram **(Figure 5.3)** illustrates the high-level boundaries of the Life Care Pharmacy system and its interactions with external entities. The system consists of two primary external entities: the User (Customer) and the Admin.

- User Inputs: The User interacts with the system by providing Registration & Login details, performing Search & Browse operations for medicines, uploading prescription images, and placing orders via Cash on Delivery (COD).
- System Outputs to User: In response, the system provides the Login Authentication Status, real-time Medicine Availability & Details, Prescription Upload Confirmation, and the final Order Confirmation along with a digital Invoice.
- Admin Interactions: The Admin manages the backend operations by inputting Admin Login Credentials, adding or updating the medicine stock, verifying/approving customer prescriptions, and updating the order delivery status.
- System Outputs to Admin: The system returns the Admin Authentication Status and provides critical dashboard monitoring data such as System Sales Reports, Customer Orders, and a list of Monitor Registered Users.

Level 1 Data Flow Diagram

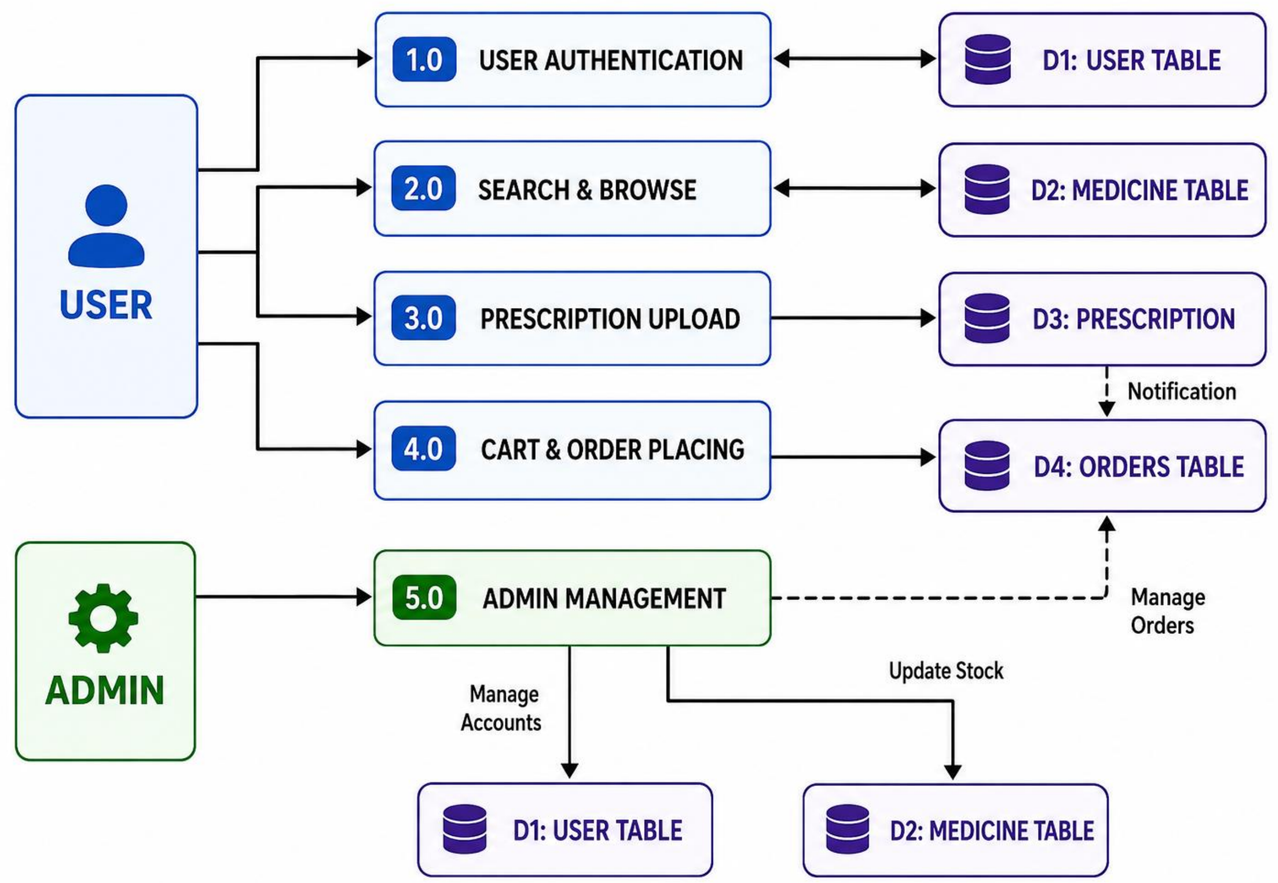


Figure 5.4: Level 1 Data Flow Diagram

Description of Level 1 DFD:

The Level 1 Data Flow Diagram **(Figure 5.4)** breaks down the main system into detailed sub-processes and shows how data is transferred between external entities and internal database tables (Data Stores). The system is divided into five core functional processes:

- User Authentication: This process handles user registration and login requests. It validates and syncs credentials with the D1: User Table.
- **2.0** Search & Browse: This process allows users to look up available medicines. It fetches real-time data from the D2: Medicine Table to display names, categories, and prices.
- **3.0** Prescription Upload: Users can upload their medical prescriptions through this interface. The data is securely saved in the D3: Prescription store, which then triggers a notification to the system for admin review.
- **4.0** Cart & Order Placing: This process manages shopping cart activities and final checkout. Once a customer places a Cash on Delivery (COD) order, the record is stored in the D4: Orders Table.
- **5.0** Admin Management: This central administrative process gives the Admin total control over the database backend. The Admin uses this module to manage user accounts (D1: User Table), update

stock levels (D2: Medicine Table), and monitor or update pending customer orders (D4: Orders Table).

CHAPTER 6

TESTING AND RESULTS

TESTING AND RESULTS

TESTING

Because it guarantees that the system operates accurately, securely, and effectively, testing is a crucial stage of website development. Various testing methods were used to confirm the medical website's performance and usefulness.

6.1 Testing Strategy

This project's testing method was created to guarantee that the medical website functions correctly, safely, and effectively for users. Various testing techniques were used to assess the system's functionality, performance, and dependability. Testing was primarily done to find mistakes, enhance user experience, and ensure that every element on the website functions properly.

Functional Testing

To confirm that all website features operate in accordance with specifications, functional testing was carried out. We thoroughly examined features including product display, add to cart, medication search, user registration, login system, and order placing.

Testing of User Interfaces

To make sure the website design is responsive and easy to use, user interface testing was done. To provide a seamless user experience, every page, button, menu, image, and form was tested on various screen sizes.

Testing databases

To ensure that all user and medication data is appropriately kept and retrieved, database testing was employed. It guaranteed accurate management of records, including customer information, orders, and medication specifics.

Testing for Security

To safeguard user and website data, security testing was carried out. To lessen unwanted access and increase system security, secure data processing techniques and login authentication were examined.

Evaluation of Performance

Performance testing aided in assessing the website's effectiveness and performance. To make sure that pages load correctly and users can access the website without any delays, the system was tested.

Testing for Compatibility

To ensure that the website works properly on various browsers and devices, including mobile phones, Chrome, and Edge, compatibility testing was carried out.

Conclusion

The medical website's functionality, usability, security, and performance were all satisfactorily confirmed using these testing techniques. The testing procedure guaranteed that users will receive dependable and effective services from the system.

6.2.Test Cases

TC ID	Test Case	Module	Steps	Expected Result	Status
TC_01	User Registration	Auth	Open site → Sign up → Submit form	Account created	Pass
TC_02	User Login	Auth	Enter email/password → Login	User logged in	Pass
TC_03	Invalid Login	Auth	Wrong credentials → Login	Error message shown	Pass
TC_04	Search Medicine	Product	Search medicine name	Results displayed	Pass
TC_05	Add to Cart	Cart	Select item → Add to cart	Item added	Pass
TC_06	Remove from Cart	Cart	Open cart → Remove item	Item removed	Pass

TC_07	Checkout Order	Order	Checkout → Enter details	Order placed	Pass
TC_08	Cash on Delivery	Payment	Select Cash on Delivery option → Confirm Order	Order Confirm	Pass
TC_09	Add Product (Admin)	Admin	Add product → Save	Product added	Pass
TC_10	Logout	Auth	Click logout	User logged out	Pass

6.3. Results and Discussion

The built medical website effectively met its primary goals of offering an online platform for pharmaceutical information, prescription management, and user-friendly pharmacy services. The system was created to improve accessibility, convenience, and communication between clients and pharmacies.

During implementation and testing, all of the website's primary modules worked properly. Users were able to explore drugs, search items, upload prescriptions, create accounts, and place orders without any significant concerns. The administrative interface also functioned well for managing products, categories, prescriptions, and customer orders. The website interface was created to be simple and responsive, making it easier to use on both desktop and mobile devices. The navigation between pages was smooth, and visitors had easy access to medical items and relevant information.

The test results revealed that:

- The website loaded pages effectively and responded appropriately to user interactions.
- The login and registration system worked properly.
- The prescription upload and order management functions worked well.
- The database stored and retrieved data accurately.
- Throughout the tests, the website maintained acceptable speed and performance.

CHAPTER 7
CONCLUSION AND FUTURE WORK

7.0 CONCLUSION AND FUTURE WORK

CONCLUSION

The built medical website successfully serves as an efficient and user-friendly online pharmacy platform. The system was developed to make pharmaceutical searches, prescription posting, and online ordering easier for users. This project successfully implemented major web development ideas such as frontend design, database administration, user authentication, and administrative control.

The website increases client convenience by providing quick access to medical products and pharmaceutical services via an internet platform. It also helps to reduce manual effort and improves communication between the pharmacy and its clients.

Overall, the project met its primary goals and demonstrated how modern web technologies may be leveraged to enhance digital healthcare services. The system can be improved in the future by include features like online payment options, delivery tracking, chatbot support, and additional security techniques.

7.1. Future Enhancements

The created medical website includes the fundamental features required for an online pharmacy system. However, additional further features can be introduced in the future to increase the system's performance, security, and usability.

Future upgrades could include:

- Credit/debit cards and digital wallets have been integrated as safe online payment alternatives.
- Customers may track their orders in real time and receive updates on their delivery status.
- Users can get help with medical and pharmacy-related questions via live chat or a chatbot.
- AI-powered pharmaceutical recommendation system provides improved user guidance.
- Mobile app development for Android and iOS platforms.
- Medicines and healthcare goods can be searched and filtered using advanced methods.
- Order confirmation and prescription update notifications are sent by email and SMS.
- Enhanced security features to safeguard user information and online transactions.
- Multilingual support makes the website more accessible to a wider audience.
- Features such as doctor consultation and appointment booking help to improve healthcare services.

7.2. Limitations

Although the built medical website successfully fulfills its primary functions, it has certain drawbacks in its current edition.

The project's key limitations are:

- The website presently supports a limited number of online pharmacy capabilities.
- The current system does not include online payment integration.
- Real-time medicine delivery tracking is not available.
- The system requires internet connectivity for effective access and performance.
- Advanced security features and data encryption methods are limited.
- The website does not offer live doctor consultation services.
- AI-based drug suggestions and smart search functionalities have not been implemented.

The present version does not have functionality for mobile applications.

Regular upgrades and maintenance may be required to improve system performance and security. The website is intended for educational and project purposes; hence some real-world pharmacy activities have been simplified.

Website link:

https://lifecarepharamcy.ourdevelopings.com/